Process Characterization Report

A-Mab Drug Substance — Cation Exchange Chromatography (Step 7) — PCR-007

Process Development / Manufacturing Science & Technology (MSAT)

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Abbreviations

AEX, anion exchange chromatography; AMV, analytical method validation; ANOVA, analysis of variance; BLA, Biologics License Application; CEX, cation exchange chromatography; Cpk, one-sided process capability index; CPP, critical process parameter; CQA, critical quality attribute; DoE, design of experiments; DS, drug substance; ELISA, enzyme-linked immunosorbent assay; GPP, general process parameter; HCP, host cell protein; HMW, high molecular weight; ICH, International Council for Harmonisation; LOQ, limit of quantitation; MSAT, Manufacturing Science and Technology; NOR, normal operating range; OD, optical density; PAR, proven acceptable range; PPQ, process performance qualification; PRESS, predicted residual error sum of squares; qPCR, quantitative polymerase chain reaction; RSD, relative standard deviation; RSM, response surface methodology; SDM, scale-down model; SEC-HPLC, size-exclusion high-performance liquid chromatography; SOP, standard operating procedure; UV, ultraviolet; WC-CPP, well-controlled critical process parameter.

Executive summary

Cation exchange chromatography is the polishing step of the A-Mab drug substance process and its principal aggregate reduction step. It uses a strong cation exchange resin in bind and elute mode, with a step elution that separates monomer from aggregate and from host cell protein, and the same operation also clears residual DNA and leached Protein A. The step forms no quality attribute of its own, so its role in the control strategy is clearance and not formation. This report presents the characterization executed under PCP-007 on a qualified scale-down model, in the lifecycle framework of (U.S. Food and Drug Administration 2011) and the enhanced development approach of (International Council for Harmonisation 2009, 2023b, 2012), with the process model and the quality attribute framework taken from the A-Mab case study (CMC Biotech Working Group 2009).

5 process parameters were characterized, 4 of them together in a multivariate design and 1 in a univariate assessment, and the outcome is 4 well-controlled critical process parameters and 1 general process parameter. No parameter of the step required the more stringent critical process parameter designation, and none is a key process parameter.

The screening design identified the active factors and the response-surface model is the predictive model used to define the operating region. Protein load is the only parameter with a significant effect on all three responses measured, load/wash conductivity governs pool host cell protein, and elution buffer pH and elution stop collect act on pool aggregate together with load. The response-surface models for pool aggregate and pool host cell protein are adequate and predictive, with coefficients of determination of 0.975 and 0.966 and no significant lack of fit against the pure error of the centre points. The model for step yield is significant overall but has a predicted coefficient of determination of only 0.20, so it is used below to describe the load trend and not to predict.

Across the whole characterized region the predicted pool aggregate stays below its acceptance limit, and at the worst corner of that region, which is protein load 40 g/L resin, load/wash conductivity 7 mS/cm, elution buffer pH 6.2, elution stop collect 1.5 OD desc., the model predicts 1.82 % HMW against a limit of 5 % HMW. At commercial scale the four attributes the step governs all meet their acceptance criteria, and the tightest of the four is host cell protein, with a mean of 15.2 ng/mg against a limit of 100 ng/mg and a Cpk of 6.14. The tightest capability of the drug substance as a whole does not arise at this step. It belongs to the cumulative parvovirus clearance claim, whose Cpk is 1.51 (PCMR-001).

Two things are not claimed here. No viral clearance credit is taken for the step, and the modular claim rests on low-pH inactivation, anion exchange and small-virus retentive filtration (PCR-006, PCR-008 and PCR-009). The step also does not bring pool host cell protein within the drug substance specification on its own. The fitted scale-down model predicts 141 ng/mg in the pool at the set-point, against a limit of 100 ng/mg and a nominal process pool of 117 ng/mg (§6), and the remaining clearance is provided by the anion exchange step (PCR-008).

2 deviations were recorded during execution, both were investigated and both were retained, and neither altered a parameter classification or the boundaries of the operating region (§13). The step is characterized over the ranges given in §4, on a scale-down model qualified against commercial-equivalent data, and the results roll up into PCMR-001.

1 Introduction

1.1 Product and unit operation

A-Mab is a humanized IgG1 monoclonal antibody produced by fed-batch mammalian cell culture and purified by a platform sequence of chromatography and filtration steps. The drug substance train runs from the production bioreactor through harvest and clarification, Protein A capture, low-pH viral inactivation, cation exchange, anion exchange, small-virus retentive filtration and ultrafiltration with diafiltration. Cation exchange is Step 7. It receives the neutralized pool of the low-pH hold and delivers the load of the anion exchange step.

The step is operated in bind and elute mode. Product binds at low conductivity and at a pH below its isoelectric point, a wash at controlled conductivity removes weakly retained impurities, and a step change to a higher pH then elutes the monomer. Pool collection starts on the ascending edge of the elution peak and stops on its descending edge at a set optical density (OD). The separation exploits differences in net charge and in charge distribution between monomer, aggregate and host cell protein. Aggregate makes more contacts with the resin than monomer does, so it is retained more strongly and elutes later, and stopping collection on the descending edge leaves most of it on the column.

Four attributes are governed here, and all four are formed before the step. At the nominal operating point the step reduces aggregate from 1.91 % HMW in the load to 0.73 % HMW in the pool, a reduction of 2.61-fold. Host cell protein is reduced 78-fold over the same run, from 9,125 ng/mg in the load to 117 ng/mg in the pool, and residual DNA and leached Protein A are cleared by 1.5 and 1 log10 respectively. These are the levels of the nominal commercial process, and the pool levels that the scale-down model of this study predicts at the same operating point are given in §6. Charge variants and glycan variants are not measured across the step, for the reason given in §2.2.

1.2 Objectives

The objectives of the study were as follows.

(i) To quantify the relationship between each process parameter of the step and the quality attributes it governs, over ranges wider than the normal operating ranges.

(ii) To define the multivariate region within which the pool attributes remain within their acceptance criteria.

(iii) To establish proven acceptable ranges for each parameter, both with the other parameters at their set-points and with the other parameters varying inside their normal operating ranges.

(iv) To classify every parameter of the step against the criteria of SOP-4001.

(v) To justify the ranges and the controls carried into Stage 2 and into the control strategy for the drug substance.

1.3 Regulatory and scientific basis

The study follows the lifecycle approach to process validation described in (U.S. Food and Drug Administration 2011). This report is a Stage 1 record for one unit operation, documenting the process design work that establishes the commercial process and its control strategy, and it precedes the process performance qualification of Stage 2. The enhanced development approach of (International Council for Harmonisation 2009) and (International Council for Harmonisation 2012) is applied throughout, and criticality is treated as a continuum and not as a binary (International Council for Harmonisation 2023b). Attributes assessed as high or very high criticality are called critical, which follows the convention of the case study (CMC Biotech Working Group 2009).

No viral clearance is claimed for this step, so the evaluation framework of (International Council for Harmonisation 2023a) is not applied here, and the steps that carry the modular clearance claim are low-pH inactivation, anion exchange and small-virus retentive filtration, whose reports are PCR-006, PCR-008 and PCR-009. Movement within the ranges established below is governed by the change management system of the pharmaceutical quality system (International Council for Harmonisation 2008).

1.4 Related documents

This report is the executed record of the plan PCP-007, its risk basis is RA-001 and its umbrella document is PCMP-001, and all three sit under the transfer plan PTP-001, which frames the technology transfer that this characterization supports. Table 1.1 gives the cross-references.

Table 1.1: Related documents in the A-Mab characterization package.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope

Document ID	Title	Relationship
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCP-007	Process Characterization Plan (Cation Exchange Chromatography (Step 7))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

2 Prior knowledge and quality risk basis

2.1 Platform and prior-product knowledge

Cation exchange polishing is a platform operation at Novacyte Biologics. It has been used at commercial scale for three previous humanized IgG1 antibodies (X-Mab, Y-Mab and Z-Mab) and throughout A-Mab clinical supply, with the resin chemistry, the buffer system and the mode of operation unchanged from those products. The feed the step receives is a Protein A pool that has passed a low-pH hold, so its composition is consistent between products, and the prior knowledge transfers to A-Mab because the binding mechanism is identical and no difference in behaviour is expected for this molecule. This characterization therefore confirms and bounds behaviour that is already understood, and it was not run to discover the mechanism.

Prior knowledge gives a directional expectation for every parameter, and the study was designed to test each one. Protein load is expected to weaken both clearances and to lower step yield, because the separation depends on the resolution between the monomer peak and the impurity peaks, and resolution falls and breakthrough begins as the column approaches its capacity. Load/wash conductivity is expected to improve host cell protein clearance, because a higher ionic strength weakens the electrostatic binding of the weakly retained host cell proteins and displaces them during the wash. Elution buffer pH is expected to raise pool aggregate, because a higher pH releases the more strongly bound species earlier in the elution. Elution stop collect is expected to raise it for a related reason, since aggregate elutes on the descending edge of the peak and a later stop collects more of that edge. Elution flow rate is expected to act through residence time and column efficiency and not through the chemistry of binding.

These expectations fix the sign of each effect, but they do not fix its size, and they say nothing about which interactions matter over the ranges the process will be operated across. Both of those questions were left to the designed experiments.

2.2 Quality attributes in scope

The step sets no quality attribute of its own. Aggregate, host cell protein and residual DNA are established upstream in the bioreactor and the harvest, and leached Protein A appears at the capture step (PCR-005). The four attributes this step controls are given in Table 2.1 with their acceptance criteria, criticality and Tool #1 score.

Table 2.1: Quality attributes controlled by the cation exchange step, with the drug substance acceptance criterion, the assigned criticality and the Tool #1 score.

CQA	Category	Acceptance	Criticality	Tool #1
Aggregates (HMW)	size variant	0–5 % HMW (SEC)	H	60
Host Cell Protein (HCP)	process impurity	0–100 ng/mg	M-H	36
Residual DNA	process impurity	0–0.001 ng/dose	VL	6
Leached Protein A	process impurity	0–5 ppm	L-M	16

Two of the four drive the study. Aggregate is of H criticality with a Tool #1 score of 60, and it is the attribute the step exists to reduce, while host cell protein is of M-H criticality with a score of 36 and is the attribute whose clearance is most sensitive to how the step is operated. Residual DNA and leached Protein A carry the lowest criticality of the four, with Tool #1 scores of 6 and 16, and RA-001 did not rank either into the multivariate scope of the study (§2.3). Both were measured on every run for information (§3.4), and neither was carried into the designed experiments as a response.

Criticality is assigned by the Tool #1 assessment of the case study, which combines the impact of an attribute on safety and efficacy with the uncertainty in that impact (CMC Biotech Working Group 2009). Attributes scored high or very high are treated as critical, and aggregate is the only attribute of this step in that class.

Charge variants and glycan variants were not measured across the step, and neither is among the attributes it controls (Table 2.1). Both are formed in the bioreactor and are reported in PCR-003. The glycan attributes are neutral variants that a charge-based separation does not resolve, so the step is not expected to move them. Acidic charge variants are a different case, since deamidation changes net charge and this step separates on charge, but they carry VL criticality with a Tool #1 score of 4 and they sit outside the scope RA-001 set for this study (§2.3).

2.3 Risk-based prioritization and parameter selection

RA-001 scored every parameter of the step before any study was executed, using risk ranking and filtering. Each parameter received one score for its potential impact on a quality attribute and one for its potential to interact with other parameters, and the combined score decided the study type. The output of that assessment is the scope of this report and not a classification, and the classifications in §9 are the outcome of the studies rather than an input to them.

4 parameters were assigned to multivariate study: protein load, load/wash conductivity, elution buffer pH and elution stop collect. All four act directly on the separation, and prior knowledge indicates that load interacts with each of the other three, which is what made a design that resolves interactions necessary. Elution flow rate was assigned to univariate assessment, because it acts on residence time and pressure drop and not on the chemistry of binding and elution. Inputs that are fixed by the resin and the column, such as resin lot, bed height and packing quality, are controlled to platform specification under SOP-2008 and were not studied.

The characterization range is wider than the normal operating range for every parameter (Table 4.1). The width was chosen so that the study bounds a region beyond the one the process will use, which is what allows an excursion to be assessed against data instead of against judgement. Protein load was taken from 10 to 40 g/L resin against a normal operating range of 18 to 32 g/L resin, which is the widest relative expansion of the four. Load is also the parameter most likely to move at commercial scale, because the titre of the harvest and the yield of the preceding steps together decide how much product reaches the column.

3 Materials and methods

3.1 Scale-down model and its qualification

The studies were executed on a qualified scale-down model of the commercial cation exchange step, operated under SOP-2010 and qualified under SOP-1001. The model holds the bed height, the linear flow velocity, the protein load per litre of resin and the wash and elution volumes in column volumes identical to the commercial step, and column efficiency was verified on plate count and peak asymmetry before each campaign. Buffer composition, pH and conductivity are the same at both scales, so the chemistry of the separation is unchanged and the principal difference between the two is column diameter.

Qualification compared the model against commercial-equivalent runs on the attributes that enter and leave the step: pool aggregate, pool host cell protein, residual DNA, leached Protein A and step yield. The comparison found no difference in performance that was significant at the level of the assays, and the qualification data set is held in the scale-down model qualification file referenced by SOP-1001 and is not reproduced in this report.

The warrant this gives is specific, and its bound matters to the claims made below. The model supports statements about how the pool attributes respond to the parameters of the step, because the separation mechanism is preserved, but it does not support statements about commercial column packing, wall effects at diameter, or the state of the resin after many cycles. The bound is the hardware, not the chemistry. Those hardware attributes are controlled by the resin life-cycle programme of SOP-2008 and are confirmed at scale during Stage 2.

The centre-point replicates of each design carry the pure-error term used for the lack-of-fit tests in §5.3. They measure how reproducible the model is at the set-point, and they include the analytical variance of the assays as well as the variability of the step itself, which is why the reproducibility figures in §5.1 are read together with the method performance in Appendix D.

3.2 Operation

Every run followed SOP-2010. The column was equilibrated in the load and wash buffer at the conductivity assigned to the run, and the neutralized low-pH inactivation pool was adjusted to the same conductivity and applied to the assigned protein load. The column was then washed with the same buffer, and the product was eluted by a step change to the elution buffer at the assigned pH. Pool collection started on the ascending edge at a fixed optical density and stopped on the descending edge at the assigned value. Flow rate was held at its set-point in every multivariate run and was varied only in the univariate assessment.

Inputs controlled to platform specification were not characterized. Resin lot, packing quality, buffer identity and the sanitization regime are identity and quality controlled under SOP-2008 and SOP-2103. The load material for both designs came from a single representative low-pH inactivation pool, so the feed state is constant across runs and contributes nothing to the estimated effects, and the consequence of that choice for the scope of the conclusions is stated in §11.

3.3 Analytical methods

Table 3.1 lists the controlled documents for the step, with the operating procedures and the validated analytical methods.

Table 3.1: Controlled documents and validated analytical methods for the cation exchange step.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2010	Operation of the Cation-Exchange Polishing Chromatography Step	SOP
SOP-2008	Chromatography Resin Life-Cycle, Packing and Sanitization	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3011	Size-Variants (SEC-HPLC)	Method validation
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation
AMV-3016	Leached Protein A by ELISA (ppm)	Method validation

Pool aggregate was measured by size-exclusion chromatography under AMV-3011, and it is the primary assay of the study. Pool host cell protein was measured by ELISA under AMV-3012, residual DNA by qPCR under AMV-3014 and leached Protein A by ELISA under AMV-3016, while step yield was calculated from the protein mass in the pool and in the load using protein concentration by ultraviolet absorbance (AMV-3019). Validated method performance is summarized in Appendix D.

Two figures from that record bear on the interpretation of the results. The host cell protein ELISA has a precision of 9.5 %RSD and accounts for a share of 0.4 of the observed variance in that response, which is why the host cell protein data are the noisiest of the three. The verification runs of DEV-007-02 used a separate verification-qualified aggregate method (AMV-3221) with a precision of 3.8 %RSD.

3.4 Sampling plan

Samples were drawn from the load and from the elution pool of every run, so that the load sample establishes the input level of each attribute for that run and the pool sample gives the response. Aggregate, host cell protein, residual DNA and leached Protein A were measured on both, and step yield was calculated from the load and pool protein concentrations and the corresponding volumes. The responses carried into the designs are pool aggregate, pool host cell protein and step yield. Residual DNA and leached Protein A were taken for information only and were not modelled as responses (§2.2), and their per-run results are held in the study record under SOP-1002 and are not reproduced here.

Centre points were replicated within each design. The screening design carried 3 centre points among its 19 runs, and the response-surface design carried 4 among its 28. Run order was randomized inside each design, so that any drift in the column or in the assays over the campaign does not align with a factor and is not mistaken for an effect.

3.5 Statistical methods

Factors were coded so that the set-point is 0 and the edges of the characterization range are ± 1, which puts the estimated effects on a common scale and makes coefficients comparable between factors with different units. All analyses were performed under SOP-1002.

The screening data were fitted by ordinary least squares with all main effects and all two-factor interactions, and an effect is reported as twice the coded coefficient, which is the change in the response across the full range of that factor. The response-surface data were fitted with the full quadratic model in the same factors. Significance was assessed at $\alpha = 0.05$ throughout.

Model adequacy was judged on the coefficient of determination and its adjusted form, on the predicted coefficient of determination computed from the PRESS statistic, on the overall F test, and on a lack-of-fit test against the pure error of the centre-point replicates. A model with a significant lack of fit is not used to define an operating region. A factor with no significant effect on any response is treated as robust over the range studied and is classified on that basis, since a null result over a range wider than the process will use is evidence and not an absence of evidence.

Commercial-scale capability was estimated by Monte-Carlo simulation of 2,000 batches, with every parameter of the train varying inside its normal operating range. Capability is reported as a one-sided Cpk against the acceptance limit, because all four attributes governed here carry an upper limit only. No acceptance threshold for Cpk is defined in this study, and the values in §8 are reported as results.

The proven acceptable ranges were computed from the fitted response-surface model by the two analyses described in §7.

4 Study design

4.1 Factors, ranges and the knowledge space

The design resolves the effects of 4 parameters on 3 responses and then describes the region those parameters define. Table 4.1 gives the set-point, the normal operating range, the characterization range, the study type and the final classification of every parameter of the step.

Table 4.1: Process parameters of the cation exchange step. The characterization range is the range studied, not a proven acceptable range; the proven acceptable ranges are computed in §7.

Parameter	Unit	Set-point	NOR	Char. range	Class	Study
Protein load	g/L resin	25	18–32	10–40	WC-CPP	multivariate
Load/Wash conductivity	mS/cm	5	4–6	3–7	WC-CPP	multivariate
Elution buffer pH	pH	6	5.85–6.15	5.8–6.2	WC-CPP	multivariate
Elution stop collect	OD desc.	1	0.7–1.3	0.5–1.5	WC-CPP	multivariate
Elution flow rate	cm/hr	200	150–250	100–300	GPP	univariate

Table 4.2: Factor legend. The letter codes are used for the terms of the effect and coefficient tables throughout.

Code	Factor	Unit	Studied in
A	Protein load	g/L resin	screening + RSM
B	Load/Wash conductivity	mS/cm	screening + RSM
C	Elution buffer pH	pH	screening + RSM
D	Elution stop collect	OD desc.	screening + RSM

The characterization range is the knowledge space of the step. It is wider than the normal operating range for every parameter, it is the region over which the models below hold, and nothing in this report supports a prediction outside it. Elution buffer pH carries the narrowest range of the four, from 5.8 to 6.2, because the elution is a step change from a released buffer and the preparation specification is tight. Elution stop collect was studied from 0.5 to 1.5 optical density units on the descending edge, a range chosen because it is the parameter that an operator or an automated trigger could most plausibly place away from target.

4.2 Screening design

The screening design was a two-level full factorial in the 4 multivariate factors with 3 centre points, 19 runs in total. A full factorial estimates every main effect and every two-factor interaction without aliasing, which a fractional design of this size could not do, and the prior expectation that load interacts with each of the other three parameters made unaliased interactions worth the extra runs. The fitted model carries 11 terms and leaves 8 residual degrees of freedom, so the fit is not saturated and the residual mean square is a usable error estimate.

The screening design carries no quadratic terms. It identifies which factors and which interactions are active and it cannot describe curvature, which is the reason a second design was executed. The coded design matrix and the measured responses are in Appendix A.

4.3 Response-surface design

The response-surface design was a face-centred central composite design in the same 4 factors, comprising 16 factorial runs, 8 axial runs on the faces of the cube and 4 centre points, 28 runs in total. The axial points sit on the faces and not beyond them, so no run exceeds the characterization range of any parameter and the model is never extrapolated within its own data. The fitted quadratic model carries 15 terms and leaves 13 residual degrees of freedom.

All 4 screening factors were carried into the response-surface design, and none could be dropped on the screening evidence, because three of them are active on pool aggregate, two are active on pool host cell protein, and the significant interactions link them to each other. The response-surface model is the predictive model of this report and the basis of the operating region in §6. The coded design matrix and the measured responses are in Appendix B.

4.4 Univariate assessment

Elution flow rate was assessed one factor at a time across 100 to 300 cm/hr, against a normal operating range of 150 to 250 cm/hr. It was kept out of the multivariate design for a mechanistic reason. Flow rate changes residence time and column efficiency, and prior platform knowledge places this separation under the control of the binding equilibrium and not of mass transfer (§2.1), so a change in flow was not expected to move the separation of monomer from aggregate and from host cell protein.

The runs were executed under SOP-2010 and analysed under SOP-1002, and their records are held in the study file and are not reproduced in this report. The assessment returned no link between flow rate and a quality attribute of the step over the range studied, which is the basis of the general process parameter classification recorded in Table 4.1. That result bounds flow rate over 100 to 300 cm/hr and no further, and the parameter stays in the knowledge space as evidence of robustness across that range.

5 Results

5.1 Centre-point performance and reproducibility

The centre points reproduce well enough for both designs to resolve the effects reported below, and they identify pool host cell protein as the least precise of the three responses. Table 5.1 and Table 5.2 give the mean, standard deviation and coefficient of variation of the centre-point replicates of each design.

Table 5.1: Centre-point reproducibility of the screening design.

Response	n	Mean	SD	%CV
Aggregates (HMW, %)	3	0.785	0.0804	10.2
Pool HCP (ng/mg)	3	155	27.2	17.6
Step yield	3	0.925	0.0131	1.42

Table 5.2: Centre-point reproducibility of the response-surface design; this is the pure-error basis of the lack-of-fit tests in §5.3.

Response	n	Mean	SD	%CV
Aggregates (HMW, %)	4	0.771	0.043	5.58
Pool HCP (ng/mg)	4	134	18.7	13.9
Step yield	4	0.918	0.0289	3.15

In the response-surface design, pool aggregate reproduced at 5.6 %CV and step yield at 3.2 %CV, while pool host cell protein reproduced at 13.9 %CV, the largest of the three. A large part of that spread is the assay. The host cell protein ELISA has a precision of 9.5 %RSD and accounts for a share of 0.4 of the observed variance in the response (Appendix D), which puts the assay component of the centre-point spread at 8.8 %CV and leaves the remainder to the step. Both parts are carried in the pure-error term used for the lack-of-fit tests in §5.3. The practical consequence is that a small change in host cell protein clearance is harder to detect at this step than a small change in aggregate clearance, and the standard errors reported in §5.2 already carry that penalty.

The screening centre points are noisier for the two impurity responses (10.2 %CV for aggregate and 17.6 %CV for host cell protein) and tighter for step yield (1.4 %CV). With 3 replicates the screening pure-error term carries 2 degrees of freedom, which is enough for a lack-of-fit check but too few to support a precision claim, whereas the response-surface design carries 4 replicates and 3 degrees of freedom and its lack-of-fit tests are reported in §5.3.

5.2 Screening: factor effects

The screening design separated the active factors from the inactive ones for both impurity responses, and the active set is different for each. Table 5.3 gives the model summary.

Table 5.3: Screening model summary. The model is the two-factor interaction model in the four multivariate factors.

Response	N	R^2	Adj. R^2	Pred. R^2	F	p-value	RMSE
Aggregates (HMW, %)	19	0.973	0.939	0.869	28.5	3.57e-05	0.096
Pool HCP (ng/mg)	19	0.936	0.857	0.467	11.8	0.000937	35.4
Step yield	19	0.924	0.829	0.498	9.74	0.00181	0.0115

All three screening models are significant overall, and their predictive quality differs by response. The predicted coefficient of determination is 0.87 for pool aggregate, which is usable, and 0.47 for pool host cell protein and 0.50 for step yield, which are not. None of the three is used to predict. The operating region below rests on the response-surface model for every response, because a two-level design carries no quadratic terms and cannot describe curvature over the region it covers.

Table 5.4: Six largest screening effects on pool aggregate, in order of absolute effect. The complete table is in Appendix C.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	0.599	0.299	0.024	12.5	1.6e-06	***
C	0.34	0.17	0.024	7.09	0.000103	***
D	0.319	0.16	0.024	6.65	0.000161	***
AC	0.196	0.0979	0.024	4.08	0.00354	**
AD	0.154	0.0772	0.024	3.22	0.0123	*
CD	0.131	0.0657	0.024	2.74	0.0256	*

Pool aggregate is driven by protein load, elution buffer pH and elution stop collect (Table 5.4). Across its characterization range protein load raises pool aggregate by 0.60 % HMW, elution buffer pH raises it by 0.34 % HMW and elution stop collect by 0.32 % HMW. All three effects are positive, all three are significant at $p < 0.001$. Every direction is the one prior knowledge predicted in §2.1. Load/wash conductivity has no effect on this response at all: its estimated effect is -0.018 % HMW at $p = 0.71$, which is smaller than the centre-point standard deviation.

Three interactions are significant on pool aggregate, and each of them pairs two of the three active factors. Protein load with elution buffer pH gives 0.20 % HMW, protein load with elution stop collect gives 0.15 % HMW, and elution buffer pH with elution stop collect gives 0.13 % HMW. All three are smaller than the main effects of the parameters that produce them, but together they are the reason a univariate range for any one of the three parameters does not describe the step, and the reason the region in §6 is multivariate.

Table 5.5: Six largest screening effects on pool host cell protein, in order of absolute effect. The complete table is in Appendix C.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
B	-127	-63.3	8.85	-7.15	9.7e-05	***
A	116	58.2	8.85	6.58	0.000173	***
AB	-56.7	-28.4	8.85	-3.21	0.0125	*
BD	39.6	19.8	8.85	2.24	0.0557	
D	-25.8	-12.9	8.85	-1.46	0.182	
AD	-24.6	-12.3	8.85	-1.39	0.202	

Pool host cell protein is driven by load/wash conductivity and protein load, and no other term reaches significance (Table 5.5). Raising conductivity across its range lowers pool host cell protein by -127 ng/mg and raising protein load raises it by 116 ng/mg, both at $p < 0.001$. The interaction between the two is significant at -57 ng/mg, and its sign says that a higher wash conductivity buys more clearance on a heavily loaded column than on a lightly loaded one. The nearest miss in the design is load/wash conductivity with elution stop collect, at 40 ng/mg and $p = 0.06$, which is the largest of the remaining estimates and is not carried forward. Elution buffer pH and elution stop collect have no significant effect on this response, which is consistent with the mechanism, because the host cell protein that survives the wash is largely removed before the elution begins and the elution conditions therefore have little left to act on.

Step yield falls with protein load and with nothing else, at an effect of -0.052 across the range and $p < 0.001$, and no other term reaches significance. The complete effect tables for all three responses are in Appendix C.

These are identification results. The magnitudes above were estimated from a design with no curvature terms, and they are refined by the response-surface model in §5.3, which is the model used for prediction, for the operating region and for the proven acceptable ranges.

5.3 Response-surface models

The response-surface models for the two impurity responses are adequate and predictive. The model for step yield is significant but not predictive, and it is used here only to describe the load trend. Table 5.6 gives the summary.

Table 5.6: Response-surface model summary. The model is the full quadratic model in the four multivariate factors.

Response	N	R^2	Adj. R^2	Pred. R^2	F	p-value	RMSE
Aggregates (HMW, %)	28	0.975	0.949	0.823	36.9	3.85e-08	0.0749
Pool HCP (ng/mg)	28	0.966	0.929	0.827	26.3	3.07e-07	16.8
Step yield	28	0.825	0.637	0.196	4.38	0.00573	0.0188

Pool aggregate and pool host cell protein have coefficients of determination of 0.975 and 0.966, adjusted values of 0.949 and 0.929, and predicted values of 0.823 and 0.827. The gap between the adjusted and the predicted value is small in both cases. Neither model is over-fitted, and both can be used to predict a mean level at a new setting inside the region. Step yield has a coefficient of determination of 0.825 and an adjusted value of 0.637, but its predicted value is 0.196. A model with a predicted coefficient of determination of that size cannot support a prediction at a new setting, so the yield result is reported below as a trend and is excluded from the design-space argument.

Lack of fit is not significant for any of the three responses, at p = 0.16, 0.68 and 0.96 against the pure error of the centre points, so the quadratic form is adequate over the region studied and no higher-order term is needed to describe the data.

Table 5.7: Response-surface coefficients for pool aggregate, in model order. Coefficients are in % HMW per coded unit.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	0.751	0.0259	29	3.36e-13	***
A	0.28	0.0176	15.9	6.77e-10	***
B	0.0276	0.0176	1.57	0.142	
C	0.172	0.0176	9.75	2.41e-07	***
D	0.157	0.0176	8.88	7.05e-07	***
AB	0.00964	0.0187	0.515	0.615	
AC	0.096	0.0187	5.13	0.000193	***
AD	0.0851	0.0187	4.54	0.000551	***
BC	-0.00211	0.0187	-0.113	0.912	
BD	0.0137	0.0187	0.733	0.477	
CD	0.0694	0.0187	3.71	0.00262	**
A^2	0.0798	0.0466	1.71	0.111	
B^2	-0.00875	0.0466	-0.188	0.854	
C^2	0.0264	0.0466	0.566	0.581	
D^2	0.0677	0.0466	1.45	0.17	

Table 5.8: Analysis of variance for the pool aggregate response-surface model, with the residual partitioned into lack of fit and pure error.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	2.89	14	0.207	36.9	3.85e-08
Residual	0.0729	13	0.0056	nan	nan
Lack of fit	0.0673	10	0.00673	3.63	0.158
Pure error	0.00556	3	0.00185	nan	nan

Table 5.7 gives the coefficient table for pool aggregate and Table 5.8 the corresponding analysis of variance. Protein load carries the largest coefficient at 0.280 % HMW per coded unit, followed by elution buffer pH at 0.172 and elution stop collect at 0.157, while load/wash conductivity is again not significant. Three two-factor interactions are significant and none of the four quadratic terms is, so the surface is close to planar with an interaction twist, and the curvature the design was executed to detect is small. The analysis of variance attributes a sum of squares of 0.067 to lack of fit on 10 degrees of freedom against 0.006 for pure error on 3, which is the least favourable of the three lack-of-fit tests and still not significant.

Table 5.9: Response-surface coefficients for pool host cell protein, in model order. Coefficients are in ng/mg per coded unit.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	141	5.81	24.3	3.21e-12	***
A	52.5	3.96	13.3	6.21e-09	***
B	-49.7	3.96	-12.6	1.21e-08	***
C	4.14	3.96	1.05	0.315	
D	-2.13	3.96	-0.538	0.6	
AB	-16.3	4.2	-3.87	0.00194	**
AC	-4.27	4.2	-1.02	0.328	
AD	-4.33	4.2	-1.03	0.321	
BC	-14	4.2	-3.33	0.00542	**
BD	0.547	4.2	0.13	0.898	
CD	-1.6	4.2	-0.381	0.71	
A^2	-16.4	10.5	-1.56	0.142	
B^2	12.9	10.5	1.24	0.238	
C^2	2.31	10.5	0.221	0.829	
D^2	10.1	10.5	0.968	0.351	

Table 5.10: Analysis of variance for the pool host cell protein response-surface model.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	104,028.0	14	7,430.6	26.3	3.07e-07
Residual	3,670.2	13	282.3	nan	nan

Source	Sum sq.	df	Mean sq.	F	p-value
Lack of fit	2,626.7	10	262.7	0.755	0.679
Pure error	1,043.5	3	347.8	nan	nan

Table 5.9 and Table 5.10 give the same for pool host cell protein. Protein load and load/wash conductivity carry coefficients of 52.5 and -49.7 ng/mg per coded unit and act in opposite directions. Two interactions are significant. Protein load with load/wash conductivity gives -16.3 ng/mg, which reproduces the screening result, while load/wash conductivity with elution buffer pH gives -14.0 ng/mg, a term that has no counterpart in the wash mechanism and did not appear in the screening analysis. The second term is retained as an empirical part of the fitted surface, and it is not read as evidence that elution pH governs host cell protein clearance. No quadratic term is significant for this response either.

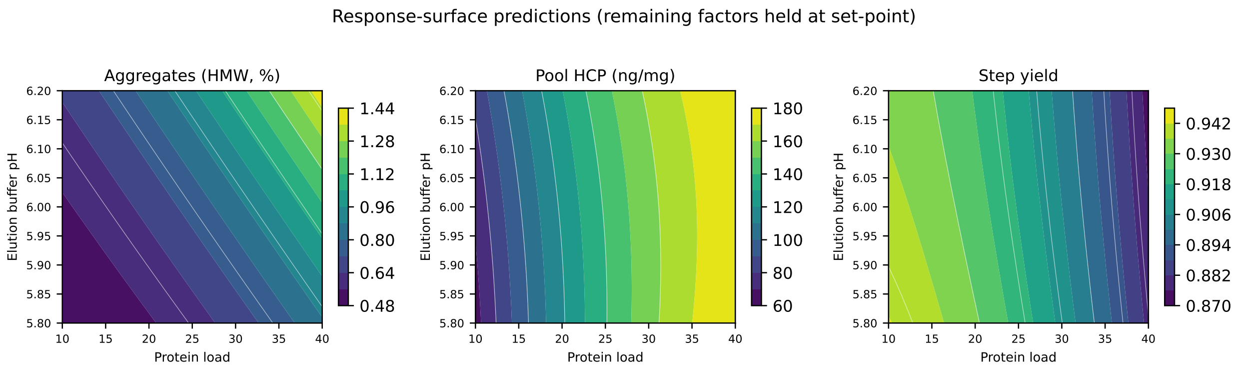

Figure 5.1: Fitted response surfaces over protein load and elution buffer pH, with load/wash conductivity and elution stop collect held at their set-points.

Figure 5.1 shows the fitted surfaces of the three responses over protein load and elution buffer pH, with the remaining two factors at their set-points. Pool aggregate rises across the panel from the low-load and low-pH corner towards the high-load and high-pH corner, and the contours fan out along the load axis, which is the graphical form of the interaction between those two parameters. Pool host cell protein varies along the load axis and is flat along the pH axis, and step yield falls along the load axis only.

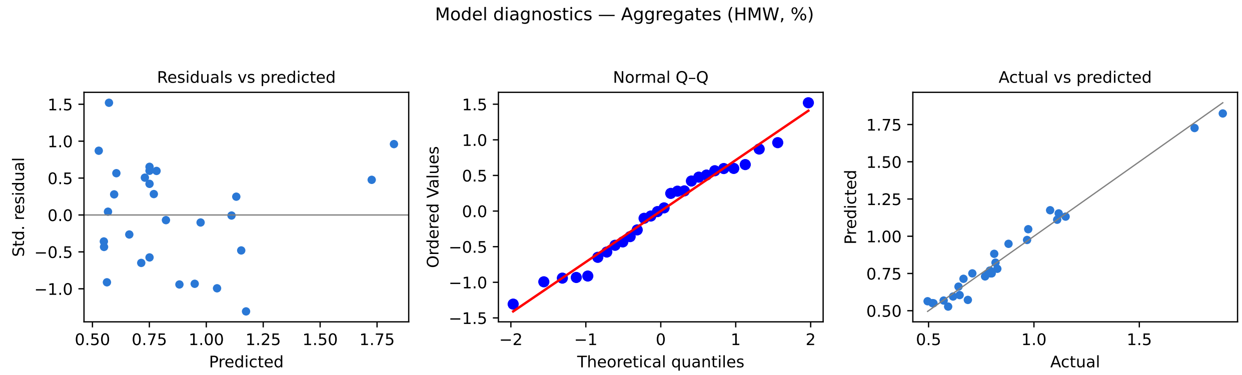

Figure 5.2: Diagnostics of the pool aggregate response-surface model: standardized residuals against predicted value, normal quantile plot, and actual against predicted.

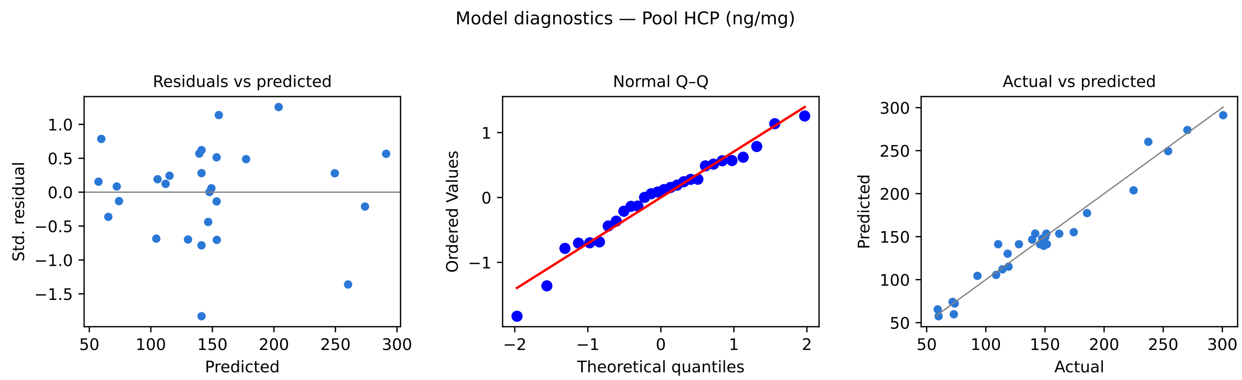

Figure 5.3: Diagnostics of the pool host cell protein response-surface model: standardized residuals against predicted value, normal quantile plot, and actual against predicted.

Figure 5.2 and Figure 5.3 give the diagnostics of the two models used for prediction. The residuals show no structure against the predicted value in either case, the normal quantile plot is close to linear, and the actual against predicted plot sits on the identity line over the whole range of the response. No run is an outlier, the largest standardized residual anywhere in the response-surface data set being 1.93. Leverage is balanced by the design itself, because the factorial and axial runs of a face-centred central composite design sit symmetrically about the centre and only the centre is replicated, so neither fit is carried by a subset of the runs.

5.4 Mechanistic interpretation

The fitted surfaces behave as the platform mechanism predicts, and the two impurity responses are governed by different parameters.

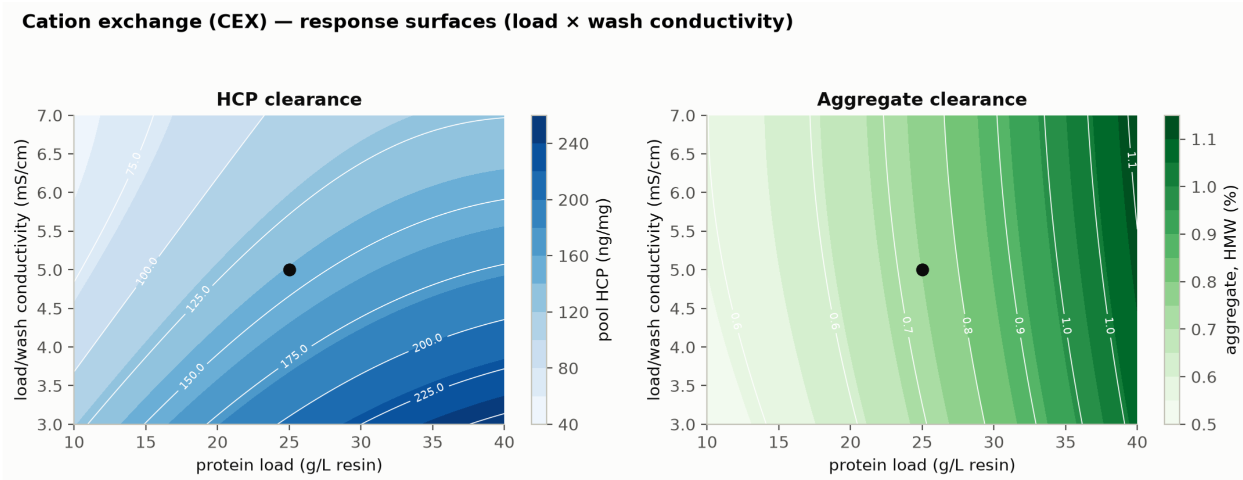

Figure 5.4: Cation exchange response surfaces for pool host cell protein and pool aggregate over protein load and load/wash conductivity, with the remaining parameters at their set-points. The marker is the set-point.

Figure 5.4 shows both impurity responses over protein load and load/wash conductivity. Protein load weakens both clearances because it reduces resolution. As the column approaches its capacity the monomer peak broadens and overlaps the more strongly retained aggregate and the more weakly retained host cell protein, so more of each ends up inside the collected pool. Load is the only parameter that carries a significant effect in all three models, and it is the largest term in two of them. That is why it carries the widest characterization range and the closest control. Between the two edges of its range the model predicts pool aggregate of 0.55 and 1.11 % HMW, and pool host cell protein of 72 and 177 ng/mg.

Load/wash conductivity acts on host cell protein alone, and it acts strongly. Host cell proteins bind through weaker and more heterogeneous electrostatic contacts than the antibody does, so raising the ionic strength of the wash displaces them while the product stays bound. Across the range of the parameter the model predicts pool host cell protein of 204 ng/mg at the low edge and 104 ng/mg at the high edge. Conductivity has no effect on pool aggregate, because aggregate and monomer differ in the number of contacts they make with the resin and not in the ionic strength at which they bind.

Elution buffer pH and elution stop collect act on aggregate through the same descending edge. A higher elution pH weakens the interaction of the strongly bound aggregate and moves it forward into the elution peak, and a later stop collect takes more of that edge into the pool. Their interaction is significant because a later stop matters more once the aggregate has already moved forward. The interaction between protein load and each elution parameter has a related origin, since a more heavily loaded column gives a broader peak and the same stop optical density then falls at a different point of the aggregate distribution.

One consequence of these mechanisms is that the step has no internal conflict between its two impurity responses. Raising the wash conductivity improves host cell protein clearance at no cost in pool aggregate, because conductivity does not appear in the aggregate model at either the screening or the response-surface stage. The only trade-off inside the step is between clearance and step yield through protein load, and it is modest: between the two edges of the load range the model predicts step yields of 0.938 and 0.876.

6 Design space

The operating region of this step is the whole characterized region as far as pool aggregate is concerned, and it is bounded for pool host cell protein by what the anion exchange step can clear. The fitted model meets the aggregate acceptance limit at 100 % of a uniform grid over the characterized region. Table 6.1 gives the prediction at the set-point, the worst case inside the normal operating ranges and the worst case at the edges of the characterization ranges, for each of the three responses.

Table 6.1: Response-surface predictions at the set-point and at the worst corner of the normal operating and characterization boxes. The worst case is the highest value for the two impurity responses and the lowest for step yield.

Response	Acceptance	At set-point	Worst case in NOR	Worst case in char. range
Aggregates (HMW, %)	≤ 5	0.751	1.27	1.82
Pool HCP (ng/mg)	≤ 100	141	209	291
Step yield	no acceptance criterion	0.917	0.892	0.865

The worst corner for pool aggregate is protein load 40 g/L resin, load/wash conductivity 7 mS/cm, elution buffer pH 6.2, elution stop collect 1.5 OD desc., where the model predicts 1.82 % HMW against an acceptance limit of 5 % HMW, a margin of 3.18 percentage points. Inside the normal operating ranges the worst case falls further, to 1.27 % HMW (Table 6.1). Aggregate does not constrain the region. The multivariate region for that attribute is the characterized region itself.

Pool host cell protein behaves differently. The model predicts 141 ng/mg at the set-point and 291 ng/mg at the worst corner, which is protein load 40 g/L resin, load/wash conductivity 3 mS/cm, elution buffer pH 6.2, elution stop collect 0.5 OD desc., against a drug substance specification of 100 ng/mg. The set-point is above that specification, and so is 82 % of the characterized region on a uniform grid of the fitted model.

Two pool levels are quoted for that operating point in this report, and they come from different models. The 141 ng/mg above is the fitted scale-down response-surface model at the set-point, while the 117 ng/mg of §1.1 is the pool of the nominal commercial process. The difference sits inside the reproducibility of the scale-down model at that point, since the response-surface centre points average 134.0 ng/mg with a standard deviation of 18.7 ng/mg (Table 5.2), which places the nominal figure 0.9 standard deviations below the scale-down centre. The region below is defined from the fitted model, so the model prediction is the figure used for it. The specification is a drug substance criterion and the pool of this step is an intermediate, so the comparison is not a failure of the step, but it does mean that no operating region can be declared here on the basis of the drug substance host cell protein limit. What leaves this step is the load of the anion exchange step, which clears host cell protein a further 6.2-fold in the nominal process (PCR-008), and the drug substance capability for the attribute is given in §8.

The principal planes of the region follow from the mechanisms in §5.4. Protein load against elution buffer pH and against elution stop collect governs pool aggregate, protein load against load/wash conductivity governs pool host cell protein, and protein load appears in both planes as the parameter that couples them. Step yield is not part of the region, because its model is not predictive (§5.3) and because yield is a process performance attribute and not a quality attribute.

The normal operating ranges sit inside the characterization ranges, which in turn sit inside the wider platform knowledge described in §2.1, and operation within the normal operating ranges is routine. Movement inside the characterization ranges but outside the normal operating ranges is supported by the data in this report for pool aggregate, which stays below its limit everywhere in the region. The same movement is not neutral for pool host cell protein, because it raises the load of the anion exchange step from 141 ng/mg at the set-point to 291 ng/mg at the worst corner, so it is assessed against the clearance credited in PCR-008 and not against this step alone. Once a design space is approved, working inside it is not a change from a regulatory filing perspective (International Council for Harmonisation 2009), and any planned move is still assessed prospectively under the pharmaceutical quality system before it is made (International Council for Harmonisation 2008). Movement outside the characterization ranges is not supported here and requires new data.

Three bounds apply to the region. It is built from mean-level predictions of a fitted model, so the assurance at its edge is lower than at its centre, in the same way that any mean-level surface gives roughly even odds at the contour it draws. It rests on a scale-down model that reproduces the separation chemistry but not commercial column packing or resin ageing (§3.1). It covers only the four parameters studied together, with the load material held at a single representative composition, so it says nothing about a load whose composition falls outside the pool specification of PCR-006.

7 Proven acceptable ranges

The acceptance criteria used here are the study-provided drug substance specifications for the two quality attributes among the measured responses: 5 % HMW for aggregate and 100 ng/mg for host cell protein, both upper limits (Table 2.1). Neither is a viral clearance attribute, so no step-level requirement had to be back-calculated from a cumulative claim. Step yield carries no acceptance criterion and is excluded from this analysis.

Two analyses were run for every combination of attribute and parameter. The first holds the other parameters at their set-points and evaluates the fitted response-surface model across the parameter's characterization range. The second varies the other parameters inside their normal operating ranges by Monte-Carlo simulation of the same model, and requires the 95% predictive interval of the attribute to stay within acceptance. The second analysis is the stricter of the two, because it asks the range to hold while the rest of the operating point moves. Table 7.1 gives both ranges for each of the eight combinations.

Table 7.1: Proven acceptable ranges for each governed attribute and parameter, from the at-set-point analysis and from the NOR-propagated Monte-Carlo analysis of the fitted response-surface model.

CQA	Parameter	Char. range	Unit	PAR (set-point)	PAR (NOR)
Aggregates (HMW, %)	Protein load	10–40	g/L resin	10–40	10–40
Aggregates (HMW, %)	Load/Wash conductivity	3–7	mS/cm	3–7	3–7
Aggregates (HMW, %)	Elution buffer pH	5.8–6.2	pH	5.8–6.2	5.8–6.2
Aggregates (HMW, %)	Elution stop collect	0.5–1.5	OD desc.	0.5–1.5	0.5–1.5
Pool HCP (ng/mg)	Protein load	10–40	g/L resin	none (set-point breaches)	none (set-point breaches)
Pool HCP (ng/mg)	Load/Wash conductivity	3–7	mS/cm	none (set-point breaches)	none (set-point breaches)
Pool HCP (ng/mg)	Elution buffer pH	5.8–6.2	pH	none (set-point breaches)	none (set-point breaches)
Pool HCP (ng/mg)	Elution stop collect	0.5–1.5	OD desc.	none (set-point breaches)	none (set-point breaches)

For pool aggregate the two analyses coincide, and the whole characterization range of every parameter is proven acceptable. Protein load is acceptable from 10 to 40 g/L resin, and the same holds for load/wash conductivity, elution buffer pH and elution stop collect across their full ranges (Table 7.1). The ranges do not narrow when the other parameters vary inside their normal operating ranges, so the result is robust to co-variation and not only to a single-parameter excursion. The highest value reached by the set-point scan of protein load is 1.11 % HMW, well below the limit.

For pool host cell protein neither analysis returns a range. The predicted pool level at the set-point is 141 ng/mg, above the drug substance limit of 100 ng/mg, so no interval containing the set-point can satisfy the criterion under either analysis and Table 7.1 records this as none for all four parameters. Measured against the drug substance specification, this step has no proven acceptable range for host cell protein. The step delivers an intermediate, the clearance that brings host cell protein to the drug substance level is shared with anion exchange (PCR-008), and what this step contributes is a clearance factor of 78-fold in the nominal process together with the parameter dependence quantified in §5.3.

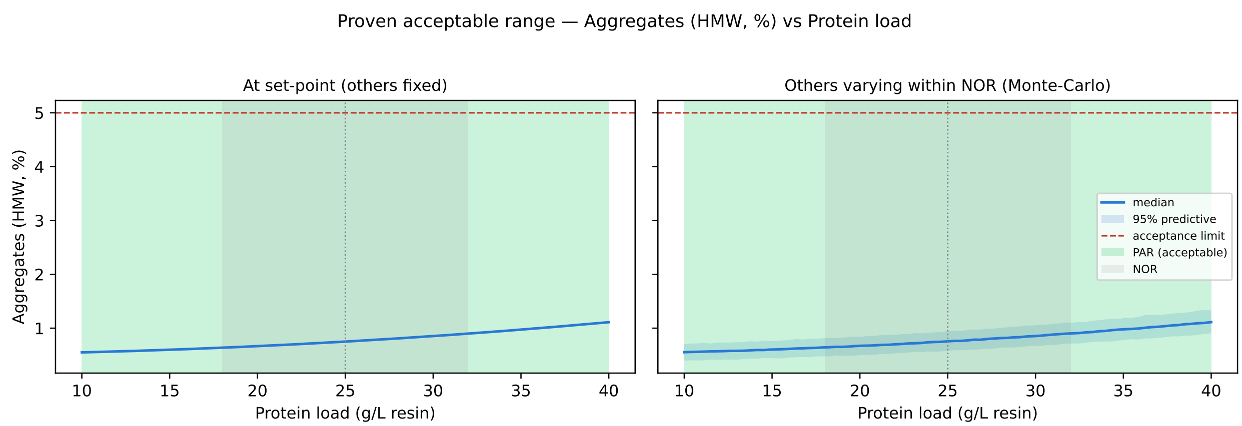

Figure 7.1: Proven acceptable range for pool aggregate against protein load, its governing parameter. Left panel, other parameters at set-point; right panel, other parameters varying within their normal operating ranges. The acceptable region is shaded green.

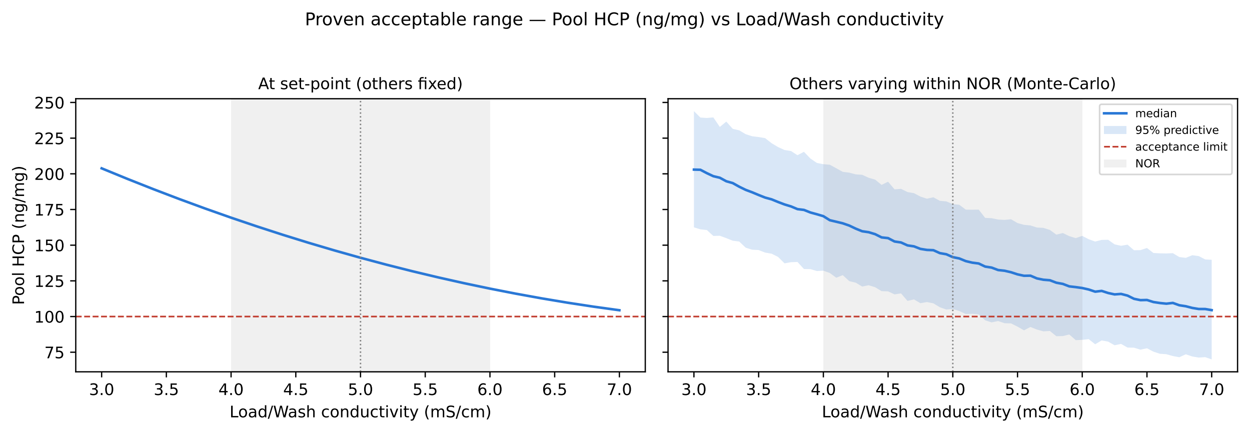

Figure 7.2: Proven acceptable range analysis for pool host cell protein against load/wash conductivity, its governing parameter. No region is shaded because the predicted pool level does not reach the drug substance acceptance limit anywhere in the range.

Figure 7.1 shows the analysis for pool aggregate against protein load, which is its governing parameter. The acceptable region covers the whole axis in both panels, and the predictive band of the right panel stays well below the limit line even at the top of the load range. Figure 7.2 shows the same for pool host cell protein against load/wash conductivity. There the predicted pool level falls as conductivity rises but does not reach the limit line, coming closest at 104 ng/mg at 7 mS/cm, so no region is shaded. That is the graphical form of the entries in Table 7.1.

The relationship between these ranges and the operating region of §6 is direct. A proven acceptable range is a univariate statement about one parameter, with the others either fixed at their set-points or varying inside their normal operating ranges, whereas the operating region is the multivariate statement. Where a parameter's whole characterization range is proven acceptable, as it is for aggregate here, the binding constraint on the operating region is not that parameter's univariate range but the worst corner of the multivariate region, which is the entry in Table 6.1.

Both analyses are bounded by the fitted model and by the characterization ranges. They predict a mean level and a predictive interval from a model estimated on 28 runs at one scale, and they say nothing about settings outside the ranges in Table 4.1. The Monte-Carlo analysis also assumes that the other parameters vary independently inside their normal operating ranges, which is the behaviour the control system is designed to produce and which is confirmed at scale during Stage 2.

8 Process capability and robustness

All four attributes the step governs meet their drug substance acceptance criteria at commercial scale, and host cell protein is the tightest of them. Table 8.1 gives the capability of each, estimated by Monte-Carlo simulation of 2,000 batches with every parameter of the train inside its normal operating range.

Table 8.1: Commercial-scale capability of the quality attributes governed by the cation exchange step. Cpk is one-sided against the upper acceptance limit.

CQA	Crit.	Spec	Acceptance	Mean ± SD	Cpk
Aggregates (HMW)	H	upper	0–5	0.753 ± 0.088	16.1
Host Cell Protein (HCP)	M-H	upper	0–100	15.2 ± 4.6	6.1
Residual DNA	VL	upper	0–0.001	0.0001 ± 0	10.8
Leached Protein A	L-M	upper	0–5	0.003 ± 0.0006	2,785.3

Aggregate has a mean of 0.75 % HMW with a standard deviation of 0.088 against an upper limit of 5 % HMW, and a Cpk of 16.1. Host cell protein has a mean of 15.2 ng/mg against 100 ng/mg and a Cpk of 6.14, which is the lowest of the four and still a wide margin, while residual DNA and leached Protein A have Cpk values of 10.8 and 2,785 and both sit far below their limits.

These capabilities belong to the drug substance and not to this step alone, and the credit divides differently for each attribute. Aggregate is the one this step effectively decides, because no later operation reduces it, and the pool level of the nominal run (0.73 % HMW) is close to the drug substance mean (0.75 % HMW). Host cell protein is shared across three steps. The Protein A step sets the level that enters (PCR-005), this step clears it 78-fold to 117 ng/mg in the nominal process, and anion exchange clears a further 6.2-fold to 18.9 ng/mg (PCR-008). Residual DNA and leached Protein A are cleared at each of the two polishing steps, and the consolidated position for both is in PCMR-001.

The tightest capability of the drug substance as a whole is none of these four. It is the cumulative parvovirus clearance, with a Cpk of 1.51, and this step takes no credit in that claim. The margins in Table 8.1 are attribute margins at this step and are not the margins of the process as a whole, and the consolidated position is in PCMR-001.

The estimate carries two bounds. It is generated from models fitted to qualified scale-down data, with parameters varying inside their normal operating ranges and not across the whole characterization ranges, so it describes routine operation and not the edge of the region. It is confirmed at commercial scale during Stage 2 (PPQ), and nothing in this report anticipates that result.

9 Parameter classification

Every parameter of the step is classified in Table 4.1, and the reason for each classification is given below. The criteria are those of SOP-4001, applied as in the case study (CMC Biotech Working Group 2009). A parameter is a critical process parameter when its variability affects a quality attribute at a level that matters to the control strategy, criticality being treated as a continuum and not as a binary (§1.3). It is called well controlled when the risk of leaving the design space during routine operation is low.

- Protein load is a well-controlled critical process parameter, because it is the only parameter with a significant effect on all three responses and the largest effect on two of them. It is well controlled because it is set by a mass calculation before the run and is verified by protein concentration (AMV-3019), so an undetected excursion is unlikely.
- Load/wash conductivity is a well-controlled critical process parameter. It is the governing parameter of pool host cell protein, with an effect of -127 ng/mg across its range, and it is measured in line and set by buffer preparation under SOP-2103, so it is readily held inside its normal operating range. Pool host cell protein is of M-H criticality and is not one of the attributes called critical in §2.2, but it is the load of the anion exchange step, so the parameter that governs it is classified on that impact.
- Elution buffer pH is a well-controlled critical process parameter, since it raises pool aggregate by 0.34 % HMW across its range and interacts with protein load. Its normal operating range is narrow and it is set by a released buffer, so the control capability is good.
- Elution stop collect is a well-controlled critical process parameter. It raises pool aggregate by 0.32 % HMW across its range and interacts with both protein load and elution buffer pH, and because it is an automated decision on the ultraviolet trace it repeats closely from run to run.
- Elution flow rate is a general process parameter. It was assessed univariately across 100 to 300 cm/hr (§4.4), that assessment returned no link to a quality attribute of the step, and the parameter stays in the knowledge space as evidence of robustness over that range.

One null result inside the multivariate set carries forward into the control strategy. Load/wash conductivity has no significant effect on pool aggregate in either design (Table 14.5), which means the conductivity range can be used to control host cell protein without a compensating cost in aggregate and is evidence that pool aggregate is robust to conductivity over the range screened. The parameter is therefore classified on its host cell protein effect alone.

No parameter of this step required the more stringent critical process parameter designation, and none is a key process parameter, so the step carries 4 well-controlled critical process parameters and 1 general process parameter.

10 Contribution to the control strategy

The step contributes five elements to the control strategy of the drug substance, and each of them is bounded by what the step actually does.

Parameter control is the first. The four multivariate parameters are held inside the normal operating ranges of Table 4.1 and are recorded in the batch record. Protein load is calculated and verified before loading, load/wash conductivity is measured in line, elution buffer pH is set by a released buffer, and elution stop collect is triggered automatically on the ultraviolet trace.

In-process monitoring is the second. Pool aggregate and pool host cell protein are measured on every batch pool (AMV-3011 and AMV-3012). The aggregate result is monitored against the drug substance specification for that attribute, because no later step reduces it, and the host cell protein result is the load level of the anion exchange step and is trended for that purpose.

Life-cycle control is the third. Resin lot, packing quality and cycle number are controlled under SOP-2008, and because this report characterizes the separation and not the resin over its lifetime, resin ageing is covered by the life-cycle programme and not by the ranges established here.

Material control is the fourth. Buffer preparation, release and expiry are controlled under SOP-2103, which is the control that DEV-007-01 failed at, and the investigation is in §13.

Release testing is the fifth. Aggregate, host cell protein, residual DNA and leached Protein A are drug substance release attributes and their acceptance criteria are in Table 2.1.

Two limits should be read with this contribution. The step does not control host cell protein to the drug substance level on its own, and the anion exchange step supplies the remaining clearance (PCR-008). The step also takes no viral clearance credit, and the modular claim rests on PCR-006, PCR-008 and PCR-009. The consolidated control strategy for the drug substance is in PCMR-001.

11 Discussion

The step is understood well enough to set its operating ranges and to defend them. Four parameters were studied together over ranges wider than they will be operated, the effects of all four are quantified, and the response-surface models for the two impurity responses are adequate and predictive. Every significant main effect has the sign that prior platform knowledge predicted in §2.1, and no unexplained factor emerged in either design. The single term without a mechanistic counterpart is the conductivity and elution pH interaction on host cell protein in the response-surface model, and it is treated as empirical (§5.3).

The scale-down model is the main assumption behind these results. It reproduces bed height, linear velocity, load per litre of resin and buffer chemistry, and it was qualified against commercial-equivalent data under SOP-1001. What it does not reproduce is a commercial column, since packing quality, wall effects at diameter and the state of the resin after many cycles all sit outside it. The consequence is bounded, because the separation mechanism is the same at both scales, so the direction and the relative size of each effect transfer even though the absolute pool levels are confirmed only during Stage 2.

2 deviations occurred and neither changed a conclusion. An expired buffer lot was used in one screening run and a column inlet temperature excursion occurred in one response-surface run, and both runs were checked against their fitted models and neither is an outlier in any response. The investigations, the impact assessments and the dispositions are in §13.

Three limitations remain. The models predict mean levels, so the reliability at the edge of the region is lower than at its centre, and the proven acceptable ranges of §7 inherit that property. No edge of the region has been run at commercial scale. The load material was a single representative pool, so the study does not bound the effect of an unusual load composition, and the attributes of that load are reported in PCR-006. Host cell protein clearance at this step cannot be assessed against a drug substance criterion at all, and the joint position with anion exchange is established in PCMR-001 rather than here.

No parameter needed the more stringent critical process parameter designation. Each of the four multivariate parameters is either calculated before the run, measured in line, or set by a released buffer or an automated trigger, so the classification stops at well controlled and the step carries no key process parameter. Should the commercial control capability for any of the four prove worse than the scale-down experience suggests, the classification is the element that would be revisited first, since the effects themselves are established.

12 Conclusions

The cation exchange step is characterized over the ranges in Table 4.1 and is robust across them for the attribute it is there to reduce. Within the characterized region the predicted pool aggregate stays below its acceptance limit at every setting, with a worst case of 1.82 % HMW against 5 % HMW. Each of the four multivariate parameters carries a proven acceptable range equal to its whole characterization range for that attribute, under both analyses of §7.

At commercial scale the four attributes the step governs meet their acceptance criteria. Host cell protein is the tightest of them, at a mean of 15.2 ng/mg against 100 ng/mg with a Cpk of 6.14. 4 parameters are classified as well-controlled critical process parameters and 1 as a general process parameter, and the step has neither a critical process parameter in the narrow sense nor a key process parameter.

Two claims are not made. The step does not deliver host cell protein at the drug substance level on its own, and no viral clearance credit is taken here. Both positions are consolidated in PCMR-001, which is also where the cumulative capability of the process is reported.

2 deviations were recorded during execution. Both were retained after investigation, and neither altered an effect estimate, a classification or a range.

13 Deviations from the plan

2 deviations were recorded during execution of PCP-007. Each was investigated, assessed for its impact on the study and dispositioned before the analysis was finalized. Both were retained, which means the affected run stayed inside the data set used for the models in §5. Table 13.1 is the register.

Table 13.1: Deviation register for the cation exchange characterization.

Deviation	Summary	Detected during	Disposition
DEV-007-01	Expired Tris load/wash buffer lot used in screening centre-point run 17	post-execution documentation review	retained
DEV-007-02	Column inlet temperature excursion during RSM run 14	equipment history review	retained

Neither deviation changed the level at which its run was executed. The expired buffer lot was used on a centre point, and the conductivity it re-tested at agrees with the centre-point level assigned to that run (§13.1). The temperature excursion acted on a variable that is not a characterized parameter of this step. Both were found after execution, by document and equipment review and not by an alarm at the time, which is itself part of the impact assessment below.

13.1 Expired buffer lot in a screening run (DEV-007-01)

An expired lot of Tris load/wash buffer (CEX) was used in screening run 17. The lot is LOT-BUF-2287 and its expiry date was 2026-02-11, which had passed when the run was executed. The deviation was found during post-execution documentation review and not at preparation, so the material control that should have caught it did not operate. Run 17 is one of the 3 screening centre points (Table 14.1), so the load/wash conductivity assigned to it is the set-point of 5 mS/cm.

The root cause is that the material expiry was not checked at buffer preparation, so the control that failed is procedural and not technical. SOP-2103 governs buffer preparation, release and expiry, and it authorizes a retained-sample re-test when an expiry deviation is found. The retained sample of the lot was re-tested under that procedure and gave a pH of 5.02 and a conductivity of 5.06 mS/cm. That conductivity agrees with the level the run was executed at, so the lot met the design setting of factor B despite its expiry, and the deviation is one of material control and not one of design execution. The corrective action is recorded in the quality system against SOP-2103 and is outside the scope of this report.

The impact on the study was assessed against the executed design. Screening run 17 was compared with the fitted screening models of §5.2, and its standardized residuals are -0.52, -0.04 and 0.65 for pool aggregate, pool host cell protein and step yield. The largest standardized residual anywhere in the screening data set is 1.99, so the run is unremarkable in all three responses and shows no sign of a degraded buffer. The run was retained, the effect estimates in §5.2 include it, and no other run in either design used this lot.

13.2 Column inlet temperature excursion in a response-surface run (DEV-007-02)

The column inlet temperature rose to 26.4 °C for 38 minutes during response-surface run 14. The excursion was found during equipment history review of the temperature-controlled chromatography chamber (EQ-CHR-118) and not by an alarm during the run. The magnitude matters because a higher column temperature can promote aggregation during elution and pool hold, so pool aggregate is the response the excursion would be expected to affect.

The root cause is a controller set-point drift on EQ-CHR-118. The chamber was in calibration at execution and its next calibration was due 2026-05-30, so the drift is a control fault and not a calibration failure, and the fault was corrected under the equipment maintenance record for EQ-CHR-118.

The run was verified before it was dispositioned. 3 verification runs were executed at the settings of run 14 and assayed for aggregate by SEC-HPLC under AMV-3221, whose precision is 3.8 %RSD, and the relative half width of the 95% confidence interval of that verification set is 9.44 %. For comparison, the effect of protein load on pool aggregate across its whole range is 78 % of the centre-point mean, so the verification bounds the run to well inside the resolution the design was built to deliver.

Run 14 was then compared with the fitted response-surface models. Its standardized residuals are -0.48, -0.44 and 0.30 for the three responses, against a largest standardized residual of 1.93 anywhere in the response-surface data set, so the run is not an outlier in any response.

The run was retained. Temperature is not a characterized parameter of this step, so the excursion changes no coefficient in Table 5.7 or Table 5.9, and it alters neither the operating region of §6 nor the classifications of §9. The residual risk is that a temperature excursion of this size is not detected in real time at commercial scale, and that risk is carried by the equipment monitoring and alarm configuration of the receiving site and not by the ranges in this report.

14 Appendices

The appendices carry the executed design matrices, the complete effect and coefficient tables, and the validated performance of the analytical methods.

14.1 Appendix A — Screening design matrix and responses

Table 14.1 gives the coded settings of the 19 screening runs and Table 14.2 the measured responses. The letter codes are those of Table 4.2.

Table 14.1: Screening design, coded settings.

Run	Type	A	B	C	D
1	factorial	-1	-1	-1	-1
2	factorial	-1	-1	-1	1
3	factorial	-1	-1	1	-1
4	factorial	-1	-1	1	1
5	factorial	-1	1	-1	-1
6	factorial	-1	1	-1	1
7	factorial	-1	1	1	-1
8	factorial	-1	1	1	1
9	factorial	1	-1	-1	-1
10	factorial	1	-1	-1	1
11	factorial	1	-1	1	-1
12	factorial	1	-1	1	1
13	factorial	1	1	-1	-1
14	factorial	1	1	-1	1
15	factorial	1	1	1	-1
16	factorial	1	1	1	1
17	center	0	0	0	0
18	center	0	0	0	0
19	center	0	0	0	0

Table 14.2: Screening design, measured responses.

Run	Type	Aggregates (HMW, %)	Pool HCP (ng/mg)	Step yield
1	factorial	0.5	136	0.936
2	factorial	0.604	114	0.962

Run	Type	Aggregates (HMW, %)	Pool HCP (ng/mg)	Step yield
3	factorial	0.548	135	0.94
4	factorial	0.818	138	0.973
5	factorial	0.452	66.2	0.934
6	factorial	0.564	61.3	0.951
7	factorial	0.579	48.8	0.946
8	factorial	0.752	67.6	0.934
9	factorial	0.792	428	0.883
10	factorial	1.1	247	0.888
11	factorial	1.14	301	0.923
12	factorial	1.79	240	0.908
13	factorial	0.807	105	0.895
14	factorial	1.04	113	0.893
15	factorial	1.12	115	0.884
16	factorial	1.83	149	0.886
17	center	0.833	153	0.929
18	center	0.829	183	0.935
19	center	0.692	129	0.91

14.2 Appendix B — Response-surface design matrix and responses

Table 14.3 gives the coded settings of the 28 response-surface runs and Table 14.4 the measured responses.

Table 14.3: Response-surface design, coded settings.

Run	Type	A	B	C	D
1	factorial	-1	-1	-1	-1
2	factorial	-1	-1	-1	1
3	factorial	-1	-1	1	-1
4	factorial	-1	-1	1	1
5	factorial	-1	1	-1	-1
6	factorial	-1	1	-1	1
7	factorial	-1	1	1	-1
8	factorial	-1	1	1	1
9	factorial	1	-1	-1	-1
10	factorial	1	-1	-1	1
11	factorial	1	-1	1	-1
12	factorial	1	-1	1	1
13	factorial	1	1	-1	-1
14	factorial	1	1	-1	1
15	factorial	1	1	1	-1
16	factorial	1	1	1	1

Run	Type	A	B	C	D
17	axial	-1	0	0	0
18	axial	1	0	0	0
19	axial	0	-1	0	0
20	axial	0	1	0	0
21	axial	0	0	-1	0
22	axial	0	0	1	0
23	axial	0	0	0	-1
24	axial	0	0	0	1
25	center	0	0	0	0
26	center	0	0	0	0
27	center	0	0	0	0
28	center	0	0	0	0

Table 14.4: Response-surface design, measured responses.

Run	Type	Aggregates (HMW, %)	Pool HCP (ng/mg)	Step yield
1	factorial	0.518	109	0.98
2	factorial	0.593	114	0.955
3	factorial	0.572	151	0.973
4	factorial	0.818	142	0.942
5	factorial	0.495	59.3	0.921
6	factorial	0.616	71.9	0.946
7	factorial	0.686	60.1	0.932
8	factorial	0.812	73	0.925
9	factorial	0.768	237	0.87
10	factorial	0.973	254	0.878
11	factorial	1.15	301	0.9
12	factorial	1.76	270	0.89
13	factorial	0.826	174	0.902
14	factorial	1.12	139	0.903
15	factorial	1.08	118	0.891
16	factorial	1.9	119	0.844
17	axial	0.523	73.7	0.935
18	axial	1.11	186	0.879
19	axial	0.666	225	0.92
20	axial	0.791	92.9	0.909
21	axial	0.647	149	0.921
22	axial	0.879	148	0.914
23	axial	0.642	162	0.935
24	axial	0.967	150	0.922
25	center	0.796	146	0.881
26	center	0.708	110	0.952
27	center	0.782	128	0.921
28	center	0.8	152	0.917

14.3 Appendix C — Full effect and coefficient tables

Table 14.5 to Table 14.7 give the complete screening effect tables for the three responses, in order of absolute effect. Table 14.8 gives the response-surface coefficient table for step yield, the response not shown in §5.3.

Table 14.5: Complete screening effects on pool aggregate.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	0.599	0.299	0.024	12.5	1.6e-06	***
C	0.34	0.17	0.024	7.09	0.000103	***
D	0.319	0.16	0.024	6.65	0.000161	***
AC	0.196	0.0979	0.024	4.08	0.00354	**
AD	0.154	0.0772	0.024	3.22	0.0123	*
CD	0.131	0.0657	0.024	2.74	0.0256	*
B	-0.0185	-0.00924	0.024	-0.385	0.71	
BC	0.0152	0.00758	0.024	0.316	0.76	
BD	-0.0131	-0.00657	0.024	-0.274	0.791	
AB	0.012	0.00602	0.024	0.251	0.808	

Table 14.6: Complete screening effects on pool host cell protein.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
B	-127	-63.3	8.85	-7.15	9.7e-05	***
A	116	58.2	8.85	6.58	0.000173	***
AB	-56.7	-28.4	8.85	-3.21	0.0125	*
BD	39.6	19.8	8.85	2.24	0.0557	
D	-25.8	-12.9	8.85	-1.46	0.182	
AD	-24.6	-12.3	8.85	-1.39	0.202	
CD	24.4	12.2	8.85	1.38	0.205	
BC	18.4	9.18	8.85	1.04	0.33	
AC	-12.5	-6.25	8.85	-0.707	0.5	
C	-9.45	-4.72	8.85	-0.534	0.608	

Table 14.7: Complete screening effects on step yield.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	-0.0518	-0.0259	0.00287	-9.04	1.79e-05	***
BC	-0.0121	-0.00607	0.00287	-2.12	0.0672	
B	-0.0111	-0.00553	0.00287	-1.93	0.0899	
AD	-0.00914	-0.00457	0.00287	-1.59	0.15	
D	0.00683	0.00342	0.00287	1.19	0.268	
C	0.00666	0.00333	0.00287	1.16	0.279	
BD	-0.00566	-0.00283	0.00287	-0.988	0.352	

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
CD	-0.00473	-0.00236	0.00287	-0.824	0.434	
AC	0.00401	0.002	0.00287	0.699	0.504	
AB	3.38e-05	1.69e-05	0.00287	0.00589	0.995	

Table 14.8: Response-surface coefficients for step yield. The model is significant overall but is not used for prediction (§5.3).

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	0.917	0.00651	141	4.36e-22	***
A	-0.0307	0.00444	-6.92	1.05e-05	***
B	-0.00757	0.00444	-1.71	0.112	
C	-0.00358	0.00444	-0.807	0.434	
D	-0.00548	0.00444	-1.24	0.239	
AB	0.00803	0.00471	1.71	0.112	
AC	0.000269	0.00471	0.0571	0.955	
AD	-0.000697	0.00471	-0.148	0.885	
BC	-0.00646	0.00471	-1.37	0.193	
BD	0.00185	0.00471	0.393	0.701	
CD	-0.00658	0.00471	-1.4	0.185	
A^2	-0.0103	0.0117	-0.882	0.394	
B^2	-0.00278	0.0117	-0.237	0.816	
C^2	0.000575	0.0117	0.049	0.962	
D^2	0.0108	0.0117	0.924	0.372	

14.4 Appendix D — Analytical methods summary

Table 14.9 gives the validated performance of the analytical methods the study relied on. Variance share is the proportion of total observed variance attributable to the method. Performance records for AMV-3011 and AMV-3014 are held with their own validation reports and are not reproduced here.

Table 14.9: Validated performance of the analytical methods used at the cation exchange step.

Method	Title	Precision (%RSD)	LOQ	Accuracy (%)	Variance share
AMV-3012	Host-Cell Protein by ELISA	9.5	1 ng/mg	95	0.4
AMV-3016	Leached Protein A by ELISA	6.5	0.25 ppm	97.4	0.41
AMV-3019	Protein Concentration by A280 (UV)	1.2	0.5 g/L	99.6	0.22

Method	Title	Precision (%RSD)	LOQ	Accuracy (%)	Variance share
AMV-3221	Aggregate by SEC-HPLC (verification-qualified)	3.8	0.2 % HMW	99.1	0.34

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